

HR Analytics – Employee Attrition Analysis

1. Introduction

Employee attrition is a critical challenge for organizations as it leads to increased recruitment costs, loss of productivity, and reduced workforce stability. This project analyzes employee data to identify the major factors contributing to attrition and provides insights to improve employee retention.

2. Objective

- Analyze employee attrition patterns.
- Identify departments and employee groups with high attrition.
- Study the impact of salary, age, education, and business travel on attrition.
- Provide recommendations to reduce employee turnover.

3. Tools Used

- Power BI
- Microsoft Excel
- Python (for data preprocessing and validation)

4. Steps Involved

1. Imported HR Analytics dataset.
2. Cleaned and prepared the data.
3. Created KPIs:
 - Employee Count
 - Attrition Count
 - Attrition Rate
 - Average Age
 - Average Salary
 - Average Tenure
4. Developed interactive Power BI dashboard.
5. Analyzed attrition across:
 - Departments
 - Salary Slabs
 - Education Fields
 - Age Groups
 - Job Roles
 - Business Travel Categories
6. Derived actionable business insights.

5. Key Findings

- Total employees analyzed: **1,470**
- Overall attrition rate: **16.08%**
- Employees aged **26–35 years** showed the highest attrition.
- Employees in lower salary bands experienced higher attrition rates.
- Sales and Research & Development departments recorded the highest employee turnover.
- Frequent business travel was associated with increased attrition.
- Employees with shorter tenure were more likely to leave the organization.

6. Attrition Prevention Suggestions

- Improve employee engagement programs.
- Review compensation for lower salary bands.
- Reduce excessive business travel where possible.
- Provide clear promotion and career growth opportunities.
- Conduct regular employee satisfaction surveys.
- Strengthen on-boarding and retention strategies for new employees.

7. Conclusion

The HR Analytics dashboard successfully identified the major factors contributing to employee attrition. The analysis revealed that age, salary, department, tenure, and business travel significantly influence employee turnover. These insights can help organizations develop effective retention strategies and improve workforce stability.

A Logistic Regression model was developed to support attrition prediction and achieved an accuracy of 84.46%, validating the patterns identified through dashboard analysis.